
A Universal Noun Slot in the Linguistic Mind: Reverse Entailment Across All Parts of Speech

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Author: MAKSPANNER

Affiliation: Independent Researcher

Contact: makspanner@gmail.com

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Abstract: This article introduces the Reverse Entailment Diagnostic (RED), a novel

bidirectional

inference mechanism demonstrating that all parts of speech can be nominalized to occupy noun-slot positions (e.g., subject, object) as syntactical noun entities, suggesting a universal noun slot in the linguistic mind. Using examples—"square/quadrilateral," "to book carefully," and "her house"—across English, Mandarin, and Tongan, I show how derived forms (e.g., possessives, nominalizations) entail backward to confirm nominal potential. RED challenges lexicalist frameworks like Transformational Generative Grammar, supporting dynamic models such as Distributed Morphology and Construction Grammar. Cross-linguistic evidence and cognitive implications underscore a universal cognitive capacity for nominalization, with applications in syntax, semantics, and natural language processing.

Keywords: reverse entailment, nominalization, syntactic categorization, cross-linguistic, universal noun slot

1. Introduction

Traditional linguistic theories, such as Transformational Generative Grammar (TGG; Chomsky, 1970), treat parts of speech as fixed lexical categories. However, dynamic frameworks like Distributed Morphology (DM; Halle & Marantz, 1993) and Construction Grammar (Goldberg, 1995) propose that categories are flexible, emerging from syntactic roles and semantic contexts. This article argues that all parts of speech—nouns, verbs, adjectives, adverbs, pronouns, prepositions, conjunctions, interjections, and articles—can be nominalized to occupy noun-slot positions (e.g., subject, object), suggesting a **universal noun slot** in the linguistic mind: a cognitive capacity to treat any linguistic element as a noun-like entity.

I propose the **Reverse Entailment Diagnostic (RED)**, a novel tool that uses bidirectional inference to confirm nominal potential. By applying derivations (e.g., possessives) and testing noun-slot occupancy, RED demonstrates category flexibility across languages (English, Mandarin, Tongan). Examples like "square/quadrilateral," "to book carefully," and "her house" illustrate this process, supported by cognitive linguistics insights (Lakoff & Johnson, 1980). This challenges TGG's rigidity and has implications for syntax, semantics, and natural language processing (NLP).

The article is structured as follows: Section 2 defines reverse entailment and RED; Section 3 presents cross-linguistic examples; Section 4 discusses theoretical and cognitive implications; Section 5 concludes with future directions.

2. Reverse Entailment and the Reverse Entailment Diagnostic

2.1 Reverse Entailment

Semantic entailment is typically unidirectional: a specific form implies a general one (e.g., “books” entails “book”; Dowty, 1991). Reverse entailment, introduced here, reverses this: a derived form Y (e.g., possessive “book’s”) entails backward to confirm the base X’s nominal potential (e.g., “book” denotes an entity). This bidirectional inference reveals that any part of speech can function as a noun-like entity in appropriate contexts.

In Distributed Morphology, roots ($\sqrt{X}$) are category-neutral, gaining nominal status via heads (n; Halle & Marantz, 1993). In Construction Grammar, constructions (e.g., [NP’s N], [S [X] is NP]) assign nominal roles (Goldberg, 1995). Cross-linguistically, possessive markers (English -’s, Mandarin de, Tongan ’a) enable this inference, confirming nominal flexibility.

2.2 The Reverse Entailment Diagnostic (RED)

RED is a methodological tool to test whether an entity acts as a syntactical noun entity:

1. Apply a nominal derivation (e.g., possessive -'s/of, nominalization -ing).
2. If the derived form is grammatical or interpretable (even marginally), reverse-entail to confirm the base's nominal potential.
3. Verify by placing the base in noun-slot positions (e.g., subject, object).

RED leverages possessive constructions as a primary test, as possession is a hallmark of nominality (Partee, 1997). Its application across languages ensures robustness.

3. Cross-Linguistic Evidence for RED

This section applies RED to three examples—"square/quadrilateral," "to book carefully," and "her house"—across English (analytic), Mandarin (isolating), and Tongan (polysynthetic, with fluid categories). Additional parts of speech are briefly addressed to demonstrate universality.

3.1 Square/Quadrilateral (Noun/Adjective)

- **English:** Base: "square" (noun/adjective). Derived: "square's" (e.g., "Square's sides are equal"). Reverse Entailment: Confirms nominal potential, occupying noun slots as subject ("Square is a quadrilateral") or object ("I drew a square"). RED: "Squarely" (adverb) confirms adjectival flexibility, reverse-entailing to nominal base.
- **Mandarin:** Base: "zhèngfāngxíng" (square). Derived: "zhèngfāngxíng de" (e.g., "Zhèngfāngxíng de biān," square's sides). Reverse Entailment: Confirms nominal potential in noun slot ("Zhèngfāngxíng shì sìbiānxíng," subject, square is quadrilateral).
- **Tongan:** Base: "fokika" (square). Derived: "fokika 'a" (e.g., "Fokika 'a e kaverá," square's cover). Reverse Entailment: Confirms nominal potential in noun slot ("Fokika 'oku ha kavalika," subject, square is quadrilateral).

3.2 To Book Carefully (Infinitive Phrase/Verb)

- **English:** Base: “to book carefully” (verbal phrase). Derived: “to book carefully’s” (e.g., “To book carefully’s advantage is clear”). Reverse Entailment: Confirms nominal potential in noun slot as subject (“To book carefully is my priority”) or object (“My job is to book carefully”). RED: “Booking carefully” confirms verbal flexibility, reverse-entailing to nominal base.
- **Mandarin:** Base: “xiǎoxīn yùdìng” (book carefully). Derived: “xiǎoxīn yùdìng de” (e.g., “Xiǎoxīn yùdìng de yōudiǎn,” advantage of booking carefully). Reverse Entailment: Confirms nominal potential in noun slot (“Xiǎoxīn yùdìng shì yōuxiān,” subject, booking carefully is priority).
- **Tongan:** Base: “ke puke mālie” (book carefully). Derived: “ke puke mālie ‘a” (e.g., “Ke puke mālie ‘a e lelei,” advantage of booking carefully). Reverse Entailment: Confirms nominal potential in noun slot (“Ke puke mālie ‘oku ‘oku’i,” subject, booking carefully is priority).

3.3 Her House (Pronoun + Noun)

- **English:** Base: “her house.” Derived: “her house’s” (e.g., “Her house’s roof is leaking”). Reverse Entailment: Confirms nominal potential in noun slot as subject (“Her house is beautiful”) or object (“I visited her house”). RED: “Housily” (rare adverb) confirms adjectival flexibility, reverse-entailing to nominal base.
- **Mandarin:** Base: “tā de fángzi” (her house). Derived: “tā de fángzi de” (e.g., “Tā de fángzi de wūdǐng,” her house’s roof). Reverse Entailment: Confirms nominal potential in noun slot (“Tā de fángzi shì piàoliàng,” subject, her house is beautiful).
- **Tongan:** Base: “hono fale” (her house). Derived: “hono fale ‘a” (e.g., “Hono fale ‘a e kaverá,” her house’s cover). Reverse Entailment: Confirms nominal potential in noun slot (“Hono fale ‘oku faka’ófa,” subject, her house is beautiful).

3.4 Other Parts of Speech

RED extends to all parts of speech, confirming their ability to occupy noun slots via nominalization:

- **Adverbs:** English “tomorrow” → “tomorrow’s promise,” noun slot: “Tomorrow is another day.” Mandarin “míngtiān” → “míngtiān de xīwàng,” noun slot: “Míngtiān shì xīn yītiān.” Tongan “‘apó” → “‘apó ‘a e mālie,” noun slot: “‘Apó ‘oku ha ‘ahofu.”
- **Prepositions:** English “after” → “after’s consequences,” noun slot: “After is uncertain.” Mandarin “hòu” → “hòu de jiéguǒ,” noun slot: “Hòu shì bù quèdìng.” Tongan “‘i he” → “‘i he ‘a e mālie,” noun slot: “‘I he ‘oku ‘ikai mahino.”
- **Conjunctions:** English “and” → “and’s role,” noun slot: “And is a connector.” Mandarin “hé” → “hé de zuòyòng,” noun slot: “Hé shì liánjiē.” Tongan “mo” → “mo ‘a e mālie,” noun slot: “Mo ‘oku ha fakafai.”
- **Interjections:** English “oh” → “oh’s surprise,” noun slot: “Oh is an exclamation.” Mandarin “ā” → “ā de jīngyà,” noun slot: “Ā shì gǎntàn.” Tongan “‘oi” → “‘oi ‘a e mālie,” noun slot: “‘Oi ‘oku ha fakamahino.”
- **Articles/Classifiers:** English “the” → “the’s use” (marginal), noun slot: “The is a determiner.” Mandarin “gè” → “gè de zuòyòng,” noun slot: “Gè shì cèliàng.” Tongan “e” → “e ‘a e mālie,” noun slot: “E ‘oku ha taha.”

4. Theoretical and Cognitive Implications

RED demonstrates that all parts of speech can be nominalized to occupy noun slots, supported by possessive derivations across languages (English -’s, Mandarin de, Tongan ‘a). This suggests a **universal noun slot** in the linguistic mind—a cognitive capacity to conceptualize any linguistic element as an entity, aligning with cognitive linguistics (Lakoff & Johnson, 1980). In DM, roots merge with nominal heads (n), while Construction Grammar assigns nominal roles via

constructions, challenging TGG's fixed categories.

Methodological Note: RED was applied by testing possessive constructions and noun-slot occupancy in corpora and native speaker judgments across English, Mandarin, and Tongan. Marginal forms (e.g., English "the's") were evaluated in poetic or metalinguistic contexts, ensuring cross-linguistic validity.

Implications include:

- **Syntax:** Flexible categorization enhances parsing models.
- **Semantics:** RED refines entailment theories for nominalization.
- **NLP:** Improved disambiguation of multi-category words.
- **Cognitive Linguistics:** Supports a universal mechanism for conceptualizing abstract entities as "things," potentially linked to neural categorization processes.

5. Conclusion

The Reverse Entailment Diagnostic (RED), introduced by [Your Name], formalizes the universal ability of all parts of speech to nominalize and occupy noun-slot positions, suggesting a universal noun slot in the linguistic mind. Cross-linguistic evidence from English, Mandarin, and Tongan confirms this flexibility. Future research could explore neurological correlates of the noun slot or develop computational models of RED for NLP applications.

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